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The Impact of Biomedical Engineers in the Efficiency of Medical Equipment Functionality and Patient Safety Outcomes

Full Name - Camila I. Santiago Quiles
Institution Name - Polytechnic University of Puerto Rico
Country - Puerto Rico
Category - Senior
Class/Year - Junior/Third Year (B.S)
Major - Biomedical Engineering
Registered Email Address - camilai.santiagoq@gmail.com
Submission Date - September 20, 2026

The Impact of Biomedical Engineers in the Efficiency of Medical Equipment Functionality and Patient Safety Outcomes

Abstract
Medical devices are central to modern patient care, yet the workforce responsible for their safe design and maintenance, biomedical engineers, remains inconsistently recognized and resourced across health care systems [1]-[3]. Existing literature describes biomedical engineer's role across the device lifecycle and documents workforce shortages, but offers limited facility evidence connecting engineering staffing to patient safety [3], [5]. This proposal outlines a mixed methods way of studying whether comparable hospitals with formal biomedical engineers show higher rates of safely functional and maintained medical equipment, and fewer equipment related safety incidents than hospitals without that integration. The study combines a structured equipment functionality audit with semi-structured staff interviews across a sample of resource limited facilities. Expected outcomes will provide facility and health policy administrators with concrete evidence to support staffing and investment decisions in biomedical engineers.

Introduction

Healthcare systems depend on an expanding base of medical technology, from patient monitors to life support equipment. The safe and efficient use of these technologies depends not only on physicians, but on a specific professional group, biomedical engineers. These are responsible for the design, evaluation and maintenance of medical devices across their operational lifecycle [1], [3]. A 2025 commentary in Lancet Global Health compared this dependence to aviation; aircraft are not considered safe strictly on a pilot's skill, but also on the strength of a recognized technical maintenance team working behind the scenes, and healthcare systems have yet to extend that recognition to biomedical engineers [5]. The World Health Organization's technical review of the biomedical engineering workforce documents inconsistency in how countries classify, train and staff this profession, with direct implications for the reliability of medical equipment in the facilities that depend on it [3]. Despite this, the existing literature on biomedical engineering's importance is mostly advocacy oriented, explaining what the profession should contribute instead of measuring what it contributes at the facility level. This proposal develops on this gap, by highlighting a research to test the relationship between formal biomedical engineering integration and two outcomes, medical equipment functionality and patient safety incident rates. This angle is timely; global health bodies have only recently begun formally documenting biomedical engineering workforce shortages at the policy level, but no study yet tests whether that shortage measurably affects equipment outcomes at the facility level, making this an unexplored and urgent research direction.

Literature Review

Saltzman's text establishes a technical view on the discipline, covering the biomechanics and biomaterials principles that support modern medical devices, and establishing biomedical engineers as the professional bridge between engineering design and clinical application [1]. Paul extends this view into applied healthcare settings , detailing the technologies used across diagnosis, treatment and rehabilitation, stating that biomedical engineering intersects nearly every stage of patient care [2]. In terms of policy, the World Health Organization's technical report frames the biomedical engineer's function across four distinct phases of the medical device lifecycle; research/innovation, evaluation, procurement and management, including safe use training for clinical staff [3]. The report also establishes cross-country evidence of workforce shortage and inconsistency, noting that many health systems have not formally recognized biomedical engineers as true professionals within the healthcare workforce [3]. Rounding out this view on policy, the equipment management literature connects staffing and training gaps to operational consequences, arguing that efficient management by biomedical engineers is a direct contributor to the quality and safety of patient care [4]. Most recently, a 2025 Lancet Global Health commentary, responding to a finding on medical oxygen during the COVID-19 pandemic, argues that no healthcare system can be considered truly effective without formally recognized biomedical engineers, calling for professional development investment and representation in health policy decision making [5]. Together, these sources establish a policy and conceptual case for the biomedical engineer's importance, but rely primarily on advocacy, description and policy analysis, rather than facility measurement. None directly test whether the integration of biomedical engineers is statistically associated with measurable equipment functionality or patient safety outcomes. This is the gap this research proposal addresses.

Research Question, Hypothesis and Variables

Research Question:To what extent does the level of formal biomedical engineering integration within a hospital's staffing structure predict or impact the functionality of its medical equipment and the frequency of equipment related, patient safety incidents?

Hypothesis: If a hospital integrates biomedical engineering staff into its equipment management, then that hospital will show a higher amount of functional medical equipment and a lower rate of incidents, regarding patient safety, than a similar hospital without such integration.

Variables: The independent variable is the level of categorized biomedical engineering integration and the dependent variables are the percentage of audited equipment found calibrated and functional, and the rate of incident reports per facility over the preceding 12 months.

Methodology

Design: A mixed methods study combining structured equipment audit (quantitative) with semi-structured interviews (qualitative).

Sampling: Sample of around 10 hospitals of comparable size and resources, brought into three groups by their existing level of biomedical engineering integration, identified through administrative contact.

Data Collection: An equipment functionality checklist applied to a set of high quality device categories at each site; recording its functional status, calibration date and completeness of maintenance records. A retrospective review of the facility's equipment incident log for the preceding 12 months. Interviews with administrators and the biomedical engineering lead, if present, covering staffing structure and maintenance practices.

Data Analysis: Quantitative data will be compared across the three integration groups using descriptive statistics and non parametric comparison to find out if differences are meaningful. Qualitative data will be analyzed using coding to identify recurring mechanisms linking staffing structure to equipment outcomes, providing an explanation behind the quantitative pattern.

Timeline: This study is expected to run along an academic term, from equipment design through audits and doing interviews at the same time, as well as a final synthesis, within a timeline that is realistic to the student.

Practicalities: A single researcher can realistically carry out the audits and interviews, and the study requires only basic statistical software, maintaining it on a reasonable budget.

Risk and Mitigation: There are two main risks to this study, the first being that inconsistent incident logs at facilities limit how reliable the equipment safety comparison is across sites. The second, and perhaps most significant, is that the small sample size causes a risk of quantitative differences not being significant enough in statistics. To reduce both of these risks we could use interview data to verify and supplement missing information, as well as frame findings as indicative, with the qualitative interviews, to provide insight that is less statistical and conclusive.

Expected Outcomes: A quantifiable comparison of equipment functionality and incident rates, paired with qualitative insight into the operational mechanisms, together forming an evidence that current literature does not yet provide [3], [5].

Conclusion

Biomedical engineers are thoroughly described in academic literature as essential to achieve an effective and safe healthcare system for patients, but evidence to prove their integration is linked to equipment and safety outcomes is still limited [3], [5]. This proposal develops an actual way, with different methods, to close that gap, giving evidence that could inform hospital administrators to better outline health policies and investment decisions, regarding these

professionals. By conjoining quantitative equipment function data and qualitative insights to the mechanisms that drive it, this proposal aims to move the idea of biomedical engineering integration from policies and principles, to practice.

References

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